

Dataset: California Housing (source: StatLib, 1990 census) — columns: longitude, latitude, housing_median_age, total_rooms, total_bedrooms, population, households, median_income, median_house_value, ocean_proximity

Questions:

1. Create an income_cat column by binning median_income with edges [0, 1.5, 3.0, 4.5, 6.0, np.inf] labeled 1–5. Using StratifiedShuffleSplit(n_splits=1, test_size=0.2, random_state=42), split the data. Report the count of each income_cat value in both the train and test sets.
2. Build a ColumnTransformer with: numeric pipeline = SimpleImputer(strategy="median") → StandardScaler(); categorical pipeline = OneHotEncoder(handle_unknown="ignore") on ocean_proximity. Fit-transform the training set and report the shape of the resulting array.
3. Using this pipeline, train LinearRegression(), DecisionTreeRegressor(random_state=42), and RandomForestRegressor(n_estimators=100, random_state=42). Using cross_val_score with cv=5 and scoring="neg_root_mean_squared_error", report the mean ± std RMSE for each model.
4. Run GridSearchCV on RandomForestRegressor over $n_estimators \in \{50, 100, 200\}$ and $max_features \in \{2, 4, 6, 8\}$ with cv=5. Report the best hyperparameter combination and its CV RMSE.
5. Evaluate the best tuned model on the held-out test set. Report Test RMSE, MAE, and R^2 .
6. Using only NumPy, implement linear regression via the Normal Equation ($\theta = (X^T X)^{-1} X^T y$) with median_income and housing_median_age (plus a bias/intercept term) as the only two features to predict median_house_value. Report your computed θ values and their % difference from sklearn.linear_model.LinearRegression fit on the same two features.
7. Plot a geographic scatter plot (longitude vs latitude, point size $\propto$ population, color $\propto$ median_house_value, alpha=0.4).

8. Plot a histogram of median_house_value.
9. Plot predicted vs. actual median_house_value on the test set (scatter plot with a red dashed diagonal reference line).
10. Plot the top-10 feature importances from the tuned Random Forest as a horizontal bar chart.